

Performance Testing

Date	03 October 2026
Team ID	17
Project Name	Fitbuddy-AI Fitness plan Generator
Department	Computer Technology
Institution / University	Madras Institute Of Technology / Anna University
Academic Year / Semester / Section	3 / 5 / Batch A
Faculty Mentor / Guide	Jayachitra V P
Team Leader	Bharath Jeyakkumar S
Sprint 1 Start / End	28-09-2026 / 28-09-2026
Sprint 2 Start / End	29-09-2026 / 30-09-2026
Sprint 3 Start / End	01-10-2026 / 02-10-2026
Team Roles	Abdul Wahith M — Backend & AI Integration; Kamalesh T — UI / Frontend Development; Bilu Besto H — Validation, Feedback & Reliability; Bharath Jeyakkumar S — Team Leader / Testing & Quality Assurance; Prasannavelan K — Database, Documentation & Demo Support
Maximum Marks	3 Marks

Test Approach

The supplied application is a synchronous local FastAPI application with AI generation as the expected latency-dominant operation. Performance testing should therefore distinguish application overhead from external Gemini latency.

Scenario	Test Setup	Measurement
1 user	Local FastAPI + mock AI	Request latency and status
10 users	Concurrent requests	Error rate and response time
25 users	Concurrent requests	Error rate and response time
Live Gemini	Real API key	End-to-end latency including model call

Evidence Status

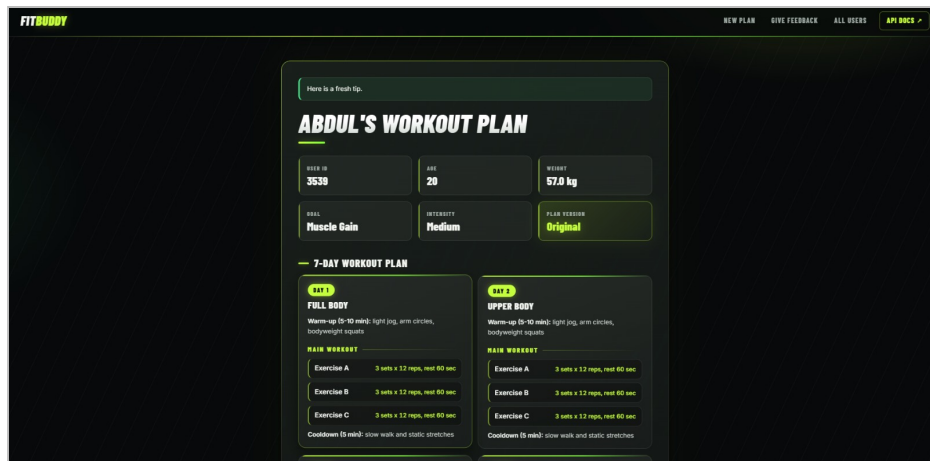
The package includes the evidence-folder convention used by the reference repository. The text evidence files are retained for the team's executed load-test outputs. Any performance values should be updated from the actual run results before final external submission if a fresh load test is performed.

Optimization opportunities

- Use an async Gemini client or background queue for long AI requests.
- Cache non-personalized nutrition requests when appropriate.
- Move SQLite to PostgreSQL for concurrent production workloads.
- Add rate limiting and observability before public deployment.

Step 5: Screenshots / Evidence

The following screenshot is visual evidence from the implemented FitBuddy application and complements the automated/text-based test evidence in 6.Project Testing/evidence/.



Generated workout plan evidence

Figure 1 — FitBuddy Generated Workout Plan Evidence. The screenshot shows the completed seven-day plan returned by the application, including the user's goal and intensity and the day-wise workout content. It demonstrates that the end-to-end generation workflow reaches a rendered result page suitable for user review.